

# **Using Data Analysis to Address Important Social Questions**

and now on to thinking  
about a social issue ...

# Consider this Statement:

## “Obesity reduces health in the community”

**Hypotheses:**

**Ha: Higher levels of obesity reduce the average level of health.** (this is what we think based on our current knowledge of obesity and health.)

Knowledge gained from where?

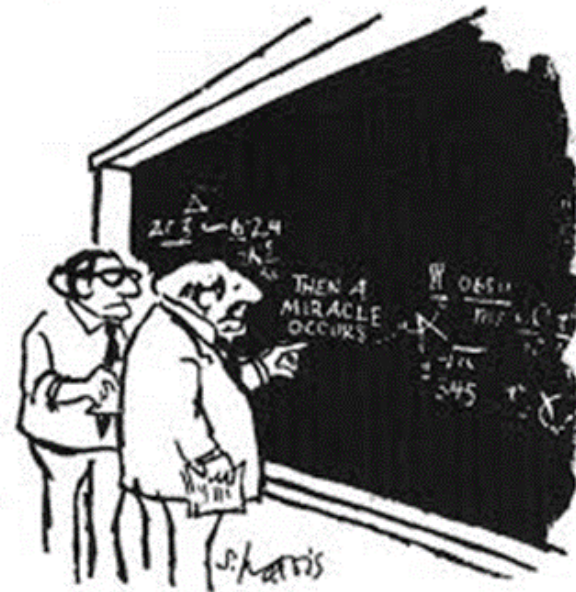
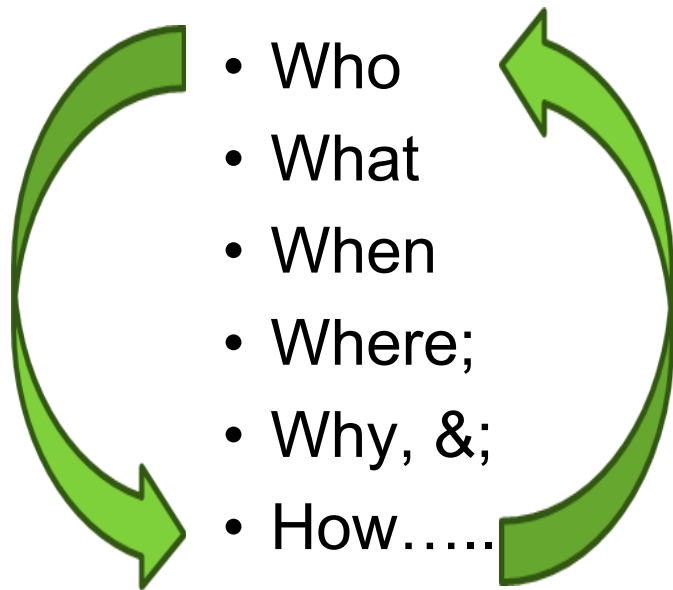
**Ho: Higher levels of obesity does not change the average level of health.**

Which hypothesis(s) do we Test and Why?

# Think about how we can test this hypothesis:

**Ho: Higher levels of obesity does not change the average level of health**

Framework for consideration you may choose to apply:



"I think you should be more explicit here in step two."

# What phenomenon are we examining?

**Ho: Higher levels of obesity does not change the average level of health**

What phenomenon of interest?

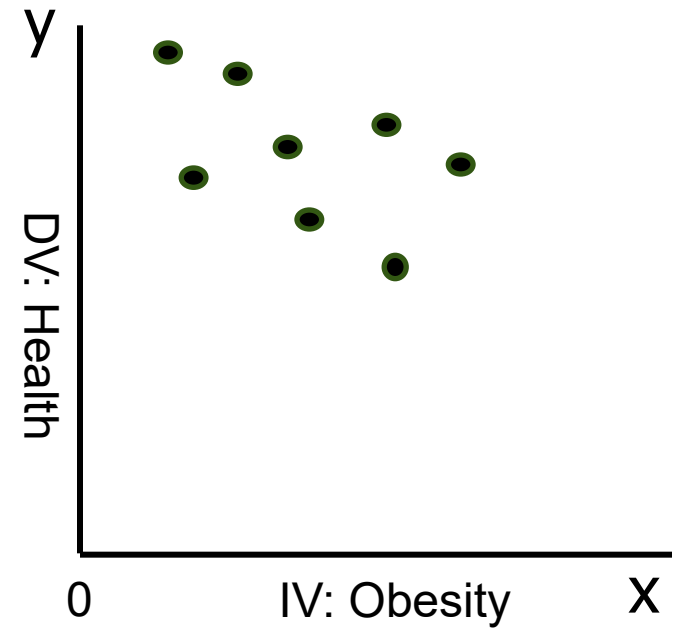
Health (dependent variable;  $y$ )

What influences that phenomenon?

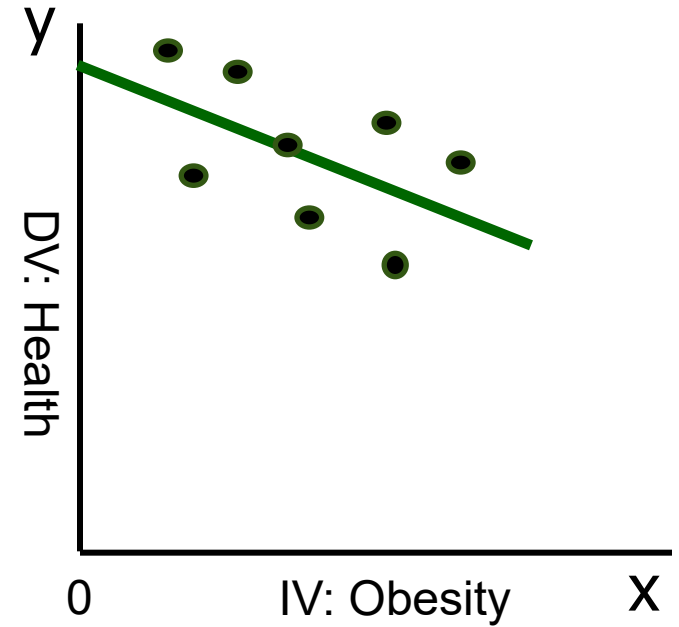
1. Obesity (independent variable;  $x$ )

**Task: Visualise this relationship and draw what you think it looks like**

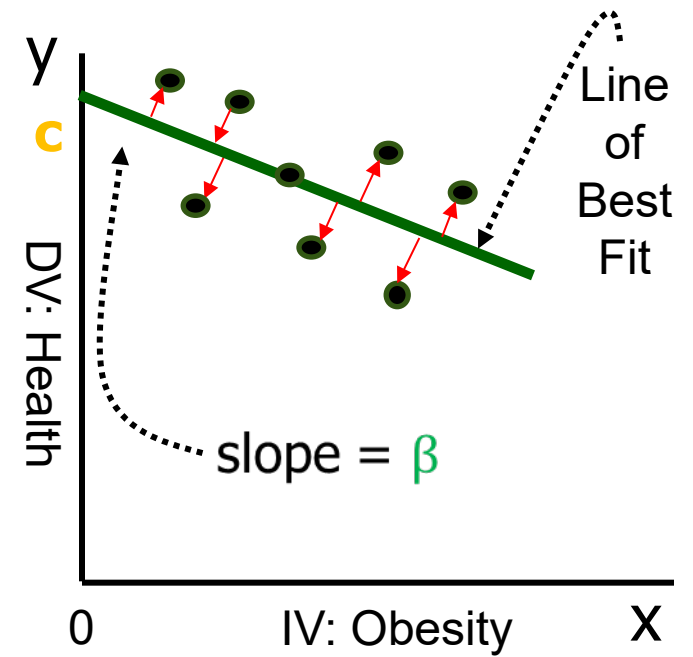
# Relationship model



# Relationship model



# Relationship model



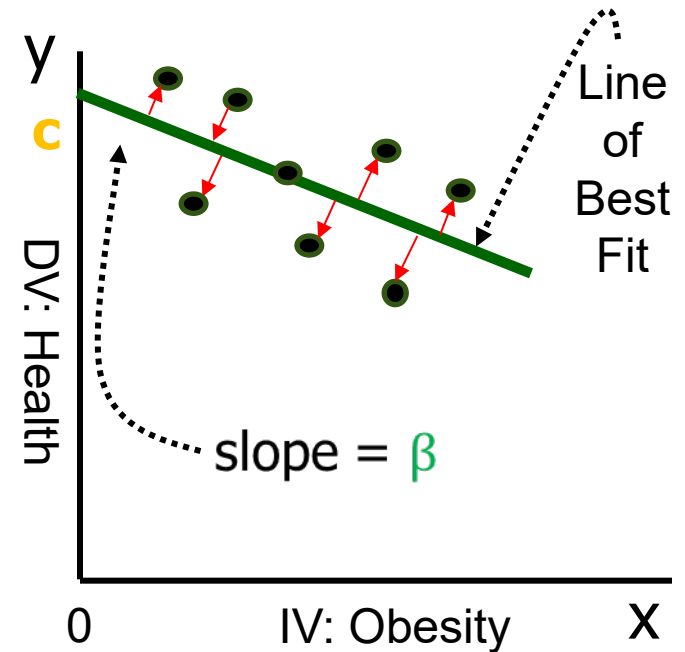
# Relationship model

**c** the intercept; the value of y when x = 0

**$\beta$**  the slope of the line; a measure of the rate of change ( $\frac{dy}{dx}$ ) in y for a unit change in x

**$\epsilon$**  that part of health that obesity does not explain; error term

Line of Best Fit: 
$$\min \sum_{n=1}^N (\epsilon^2)$$



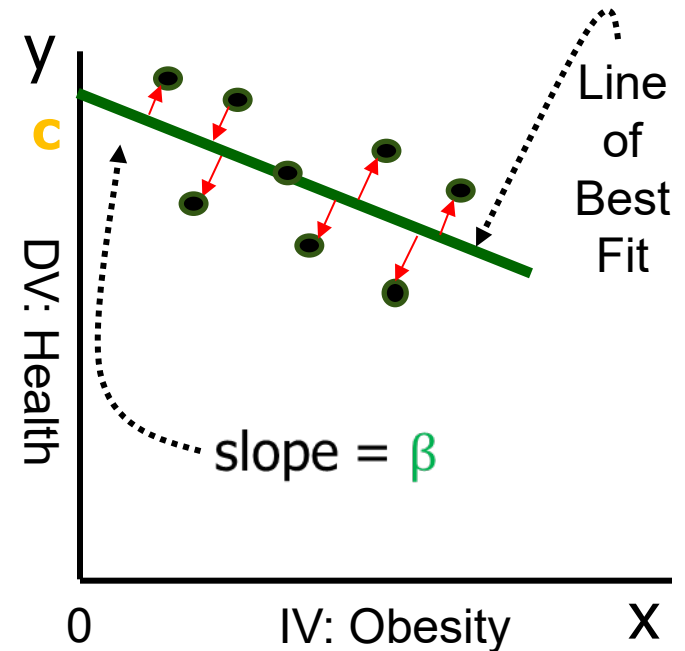
# Linear Relationship model

$c$  the intercept; the value of  $y$  when  $x = 0$

$\beta$  the slope of the line; a measure of the rate of change ( $\frac{dy}{dx}$ ) in  $y$  for a unit change in  $x$

$\epsilon$  that part of health that obesity does not explain; error term

Line of Best Fit:  $\min \sum_{n=1}^N (\epsilon^2)$



Equation of the line:  $y=f(x)$ ;  $y = c - \beta x + \epsilon$ ;  $\{x_1, x_2, x_3 \dots x_N\}$

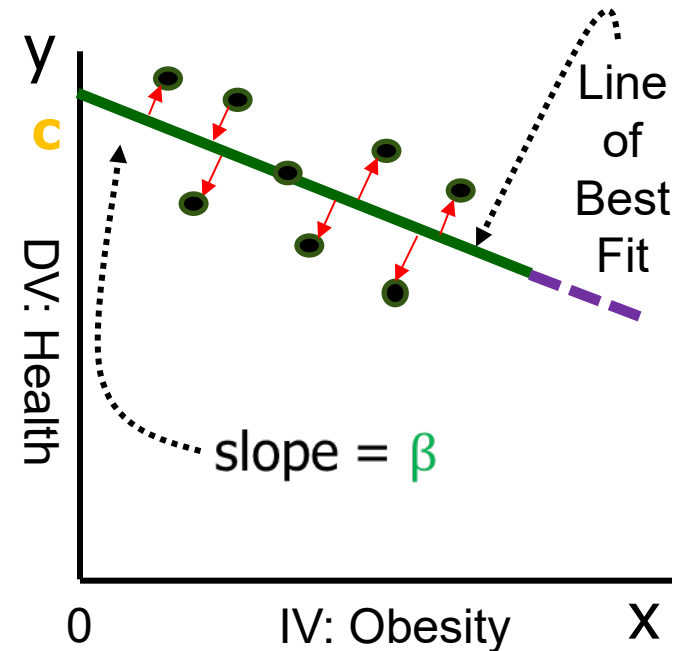
# Linear model (Ordinary Least Squares)

**c** the intercept; the value of y when x = 0

**β** the slope of the line; a measure of the rate of change ( $\frac{dy}{dx}$ ) in y for a unit change in x

**ε** that part of health that obesity does not explain; error term

Line of Best Fit: 
$$\min \sum_{n=1}^N (\epsilon^2)$$



*Equation of the line:  $y=f(x)$ ;  $y = c - \beta x + \epsilon$ ;  $\{x_1, x_2, x_3 \dots x_N\}$*

**Prediction:** If we do not have data past a certain value of obesity (say 30), how can we predict the change in average health (y) if obesity (x) was to rise to 32?

**Apply the equation of the line:**  $\beta = -0.4$ ;  $c = 4.5$ :  $4.5 + (-0.04 \cdot 32) = 3.22$

**Task: Predict health when obesity=50**

# What else has an effect on Health?

**Ho: Higher levels of obesity does not change the average level of health**

What phenomenon of interest?

Health (dependent variable;  $y$ )

What influences that phenomenon?

1. Obesity (independent variable;  $x$ )
2. Etc.
3. Etc.

**What else could have an effect on health? (control variables)**

# Task: What else has an effect on Health?

Hint: How can I find out what else has an effect on health?

Your List of variables?

# Time to collect some data

**Ho: Higher levels of obesity does not change the average level of health**

*For Who:* Australians; the population for our study.

**Task: How would you rewrite the hypotheses now that we have decided to look at Australians?**

**Ha: Higher levels of obesity in Australians reduces the average level of health of Australians.**

**Ho: Higher levels of obesity in Australians does not change the average level of health of Australians.**

# Population v Sample

**Population**

Measures used to describe a population are called **parameters**

**All Australians on  
the electoral role**

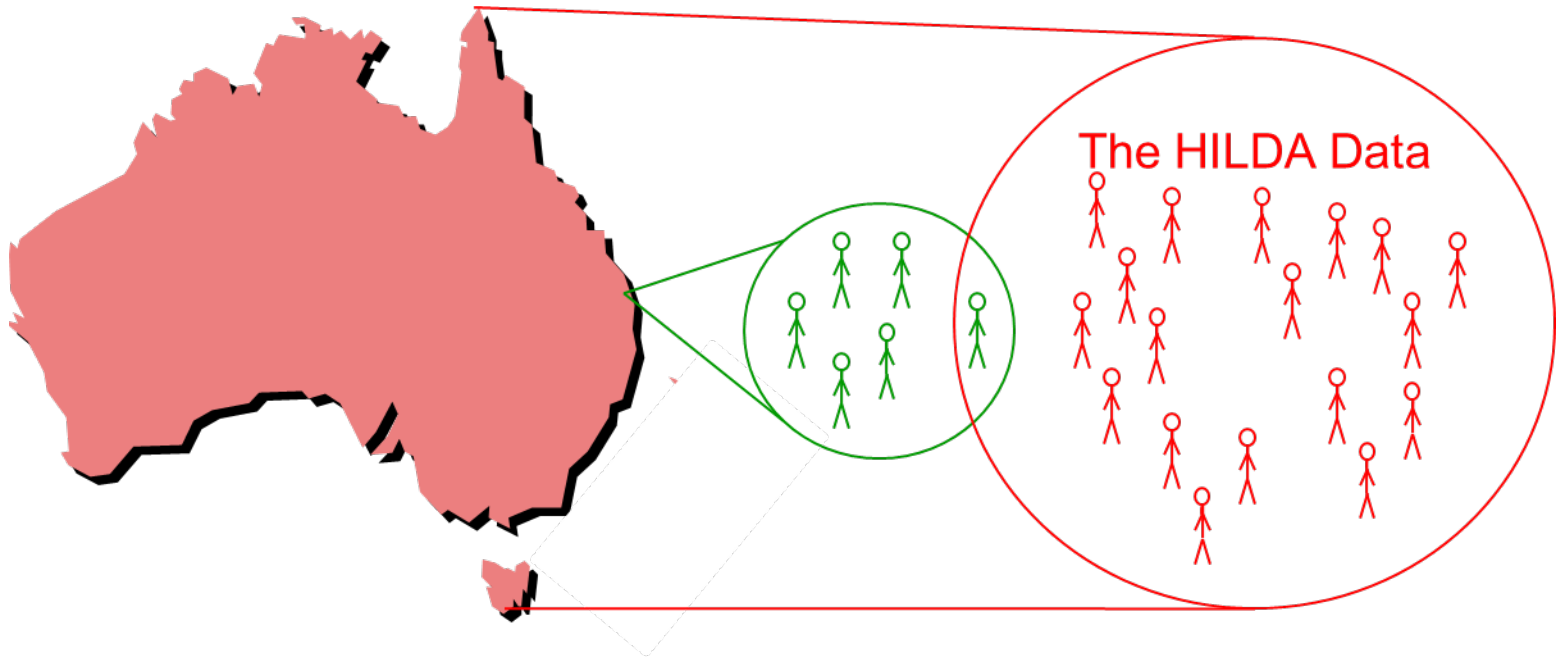
**Australians who  
respond to a survey**

**Representative  
Sample**

Measures computed from sample data are called **statistics**

# Collect Data from Who?

- **WHO** are we interested in (Australian population)?
- Do we want to collect data from Australian: **Individuals** (*i*), **groups** (*g*)? (unit of analysis)?
- Which people in the Australian population do we ask to complete a survey (random sampling)?



[Example: HILDA for Data Users \(unimelb.edu.au\)](http://unimelb.edu.au)

# Time to collect data

- At one point in time (cross-sectional data);
- Over a Number of Periods ( $t$ ) (time-series data);
- Pooled, a combination of Time Series and Cross-sectional, or;
- Something else?

**Task: How often do we want to collect the data and why?**

# Types of Data

- **What** type of data would inform us about the DV?
- **What** type of data would inform us about the IVs?

**Task 4: Consider the above**

# Time to collect data

What type of data would inform us about the DV?

Health: self-reported general health factor from the SF-36 international health self-reported health question:

**PART A: GENERAL HEALTH AND WELL-BEING**  
(SF-36 Health Survey)

*This first set of questions seeks your views about your health, how you feel and how well you are able to do your usual activities.*

*Please take the time to read and answer each question carefully by crossing the box corresponding to your response. If you are unsure about how to answer a question, please give the best answer you can.*

**A1** In general, would you say your health is:

(Cross ☒ **ONE** box)

|                                                 |                                                 |                                            |                                            |                                            |
|-------------------------------------------------|-------------------------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|
| <input type="checkbox"/> <sub>1</sub> Excellent | <input type="checkbox"/> <sub>2</sub> Very good | <input type="checkbox"/> <sub>3</sub> Good | <input type="checkbox"/> <sub>4</sub> Fair | <input type="checkbox"/> <sub>5</sub> Poor |
|-------------------------------------------------|-------------------------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|

sgh1

Scale: 1 (min) to 5 (max); nominal, discrete ( 1,2,3,4,5); range = 5; reverse coded; rescale

Is the scale Cardinal, or;  
Ordinal?

0 1 2 3 4 5  
0 1 2 3 4 5

**Task: What difference would ordinal/cardinal data make to your visualisation of the obesity - health relationship?**

# Time to collect data

What type of data would inform us about the IV?

Obesity: Body Mass Index (BMI); *f* (self-reported weight [kg]; height [metres<sup>2</sup>])

B8 How tall are you (without shoes)?

You only need to provide an answer in either centimetres (cms) or in feet / inches.

 cms

OR

 feet  inches  
(Note: There are 12 inches in a foot)

sbmhtcm  
sbmhtft  
sbmhtin

B9 What is your current weight?

You only need to provide an answer in either kilograms (kgs) or in Stones / pounds.

 kgs

OR

 Stones  pounds  
(Note: There are 14 pounds in a stone)

sbmwtkg  
sbmwkst  
sbmwtlb

Scale (kg): numeric, continuous; range  $0 = \infty$

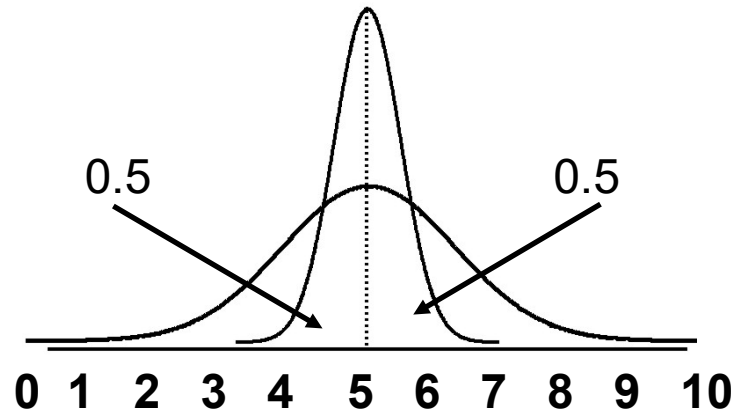
$$\text{BMI} = \frac{\text{weight (kg)}}{\text{height (m}^2\text{)}}$$

**Task: Consider the accuracy of self-reported height and weight, and the constructed BMI factor?**

# Other types of data sources & scales?

**Task: Consider other DVs; IVs; data types and scales; list them.**

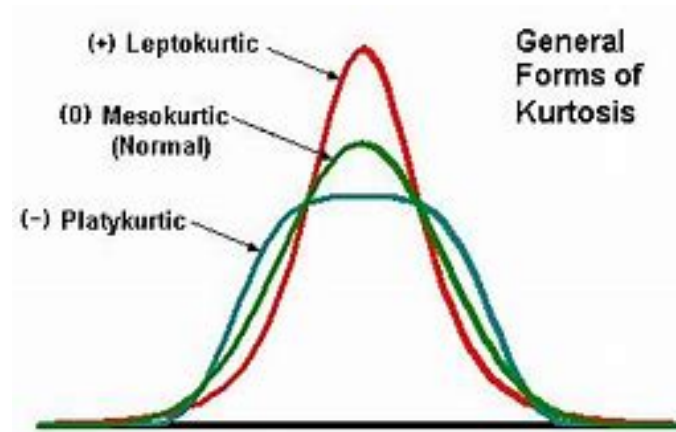
# Normal Distribution of a Variable



Task: What happens to the Distribution of a variable as the Sample Size changes?

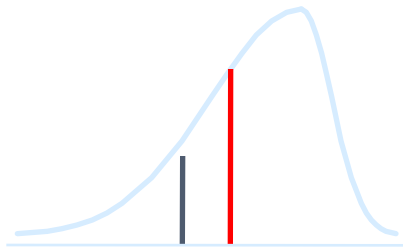
# Shape of a Distribution

Measures of shape describe how data are distributed: symmetric or skewed; kurtotic; outliers



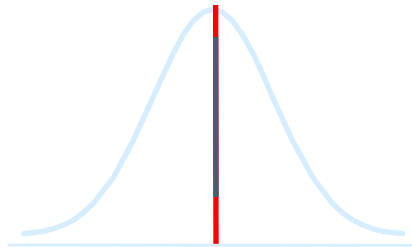
## Left-Skewed

Mean < **Median**



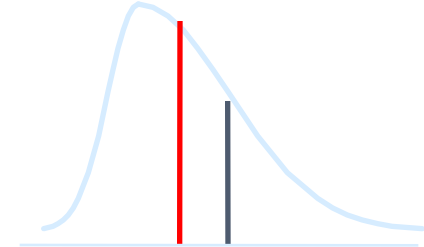
## Symmetric

Mean = **Median**

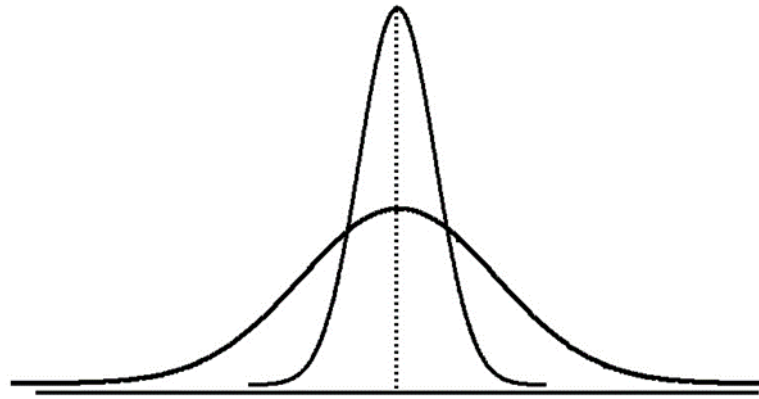


## Right-Skewed

**Median** < Mean



# Distribution of the Health & BMI Variables

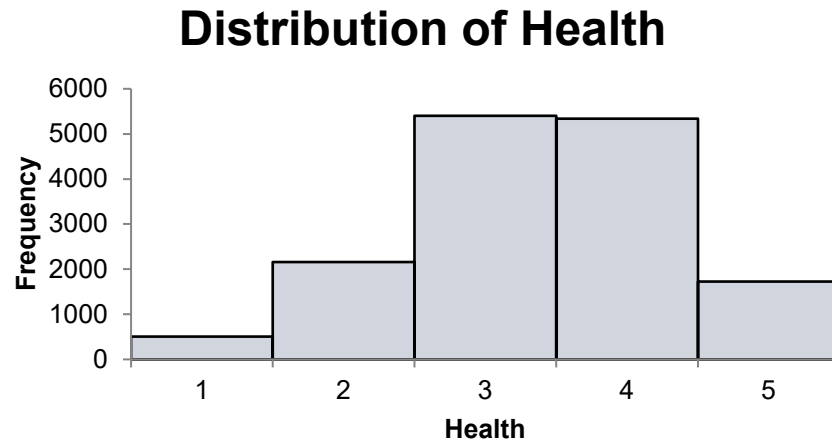


**Task: Illustrate the distribution of: BMI; Health**

# Excel

## Pivot Table

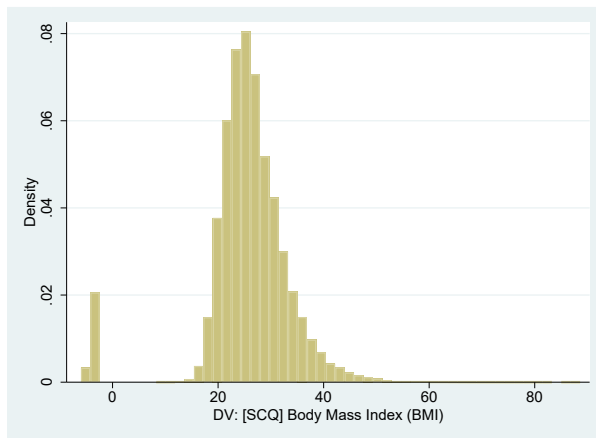
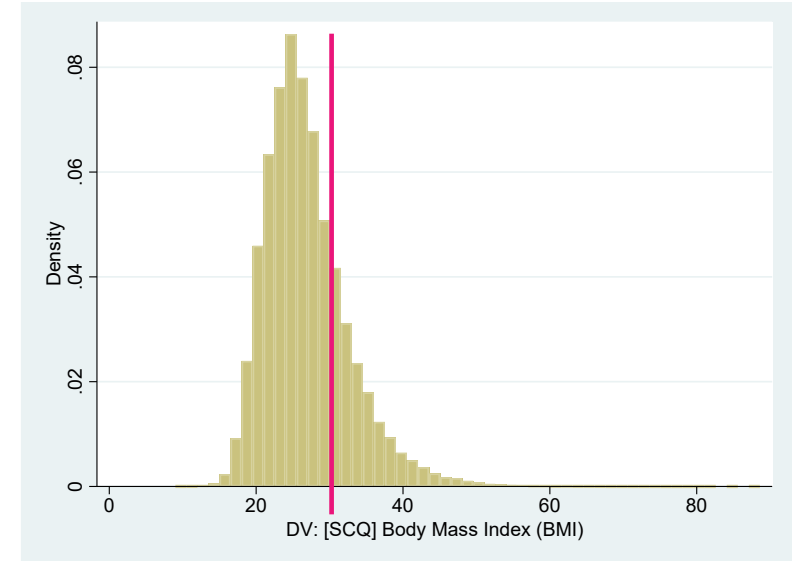
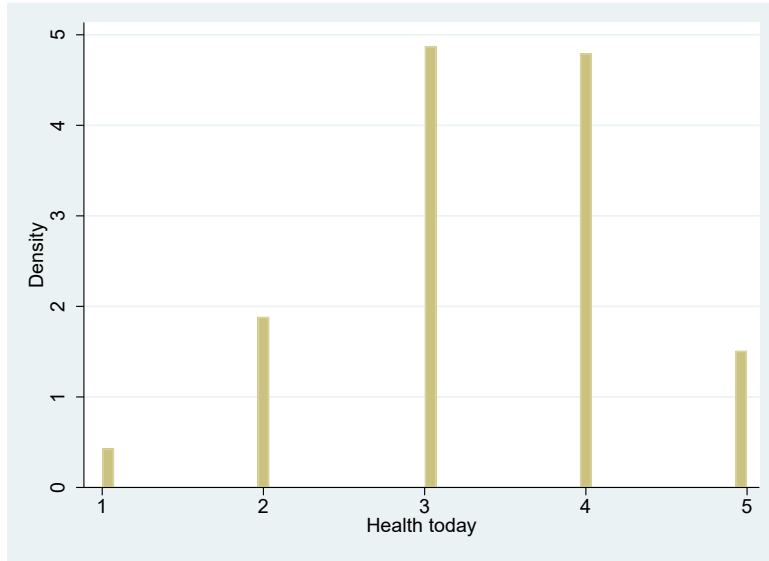
### Excel Data Analysis Package Pivot Table



**Task: Illustrate the distribution of: a. BMI; b. Health**

# Distribution of Health & Obesity

**obese** defined as a BMI 30 or greater (Blyth, Frijters & Beaton, 2023)

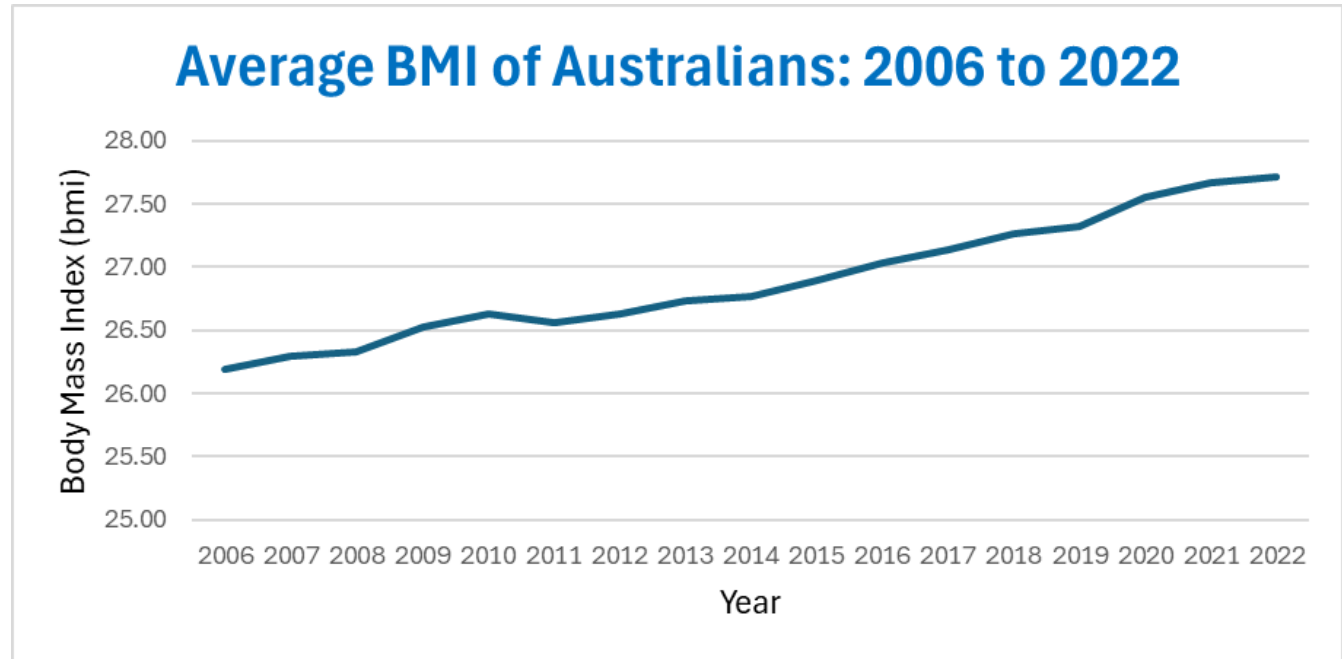


| DV: | [SCQ] Body Mass Index (BMI) |       | Freq. | Percent |
|-----|-----------------------------|-------|-------|---------|
|     |                             | Cum.  |       |         |
|     | [-6.0] Implausible value    | 1,427 | 0.59  | 0.59    |
|     | [-4.0] Refused/Not stated   | 8,863 | 3.68  | 4.27    |
| 9   |                             | 1     | 0.00  | 4.27    |
| 9.6 |                             | 1     | 0.00  | 4.27    |

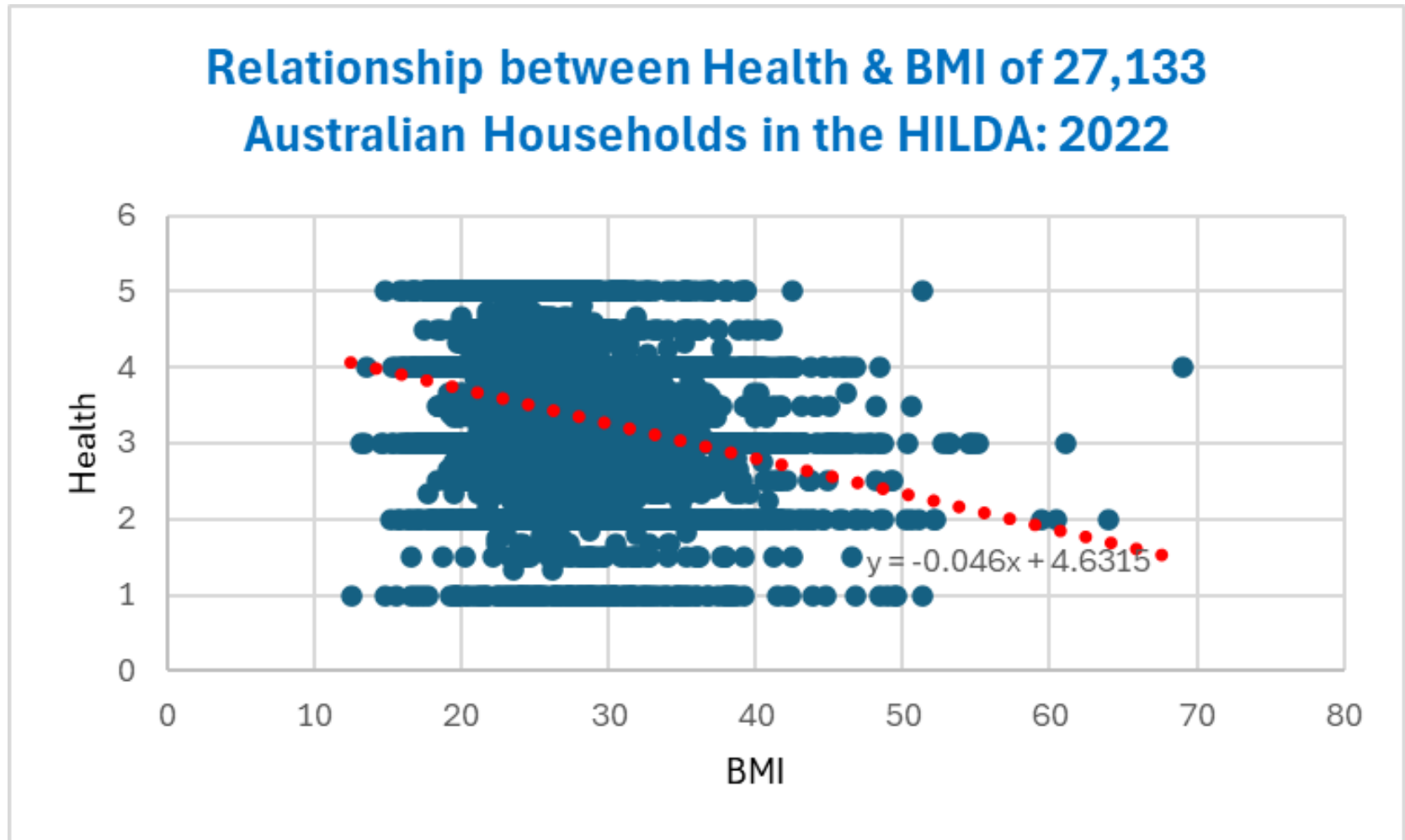
Distribution of a Variable is Determined by its Values.

## 6. Graphically represent the Health – BMI relationship and interpret: Time-series

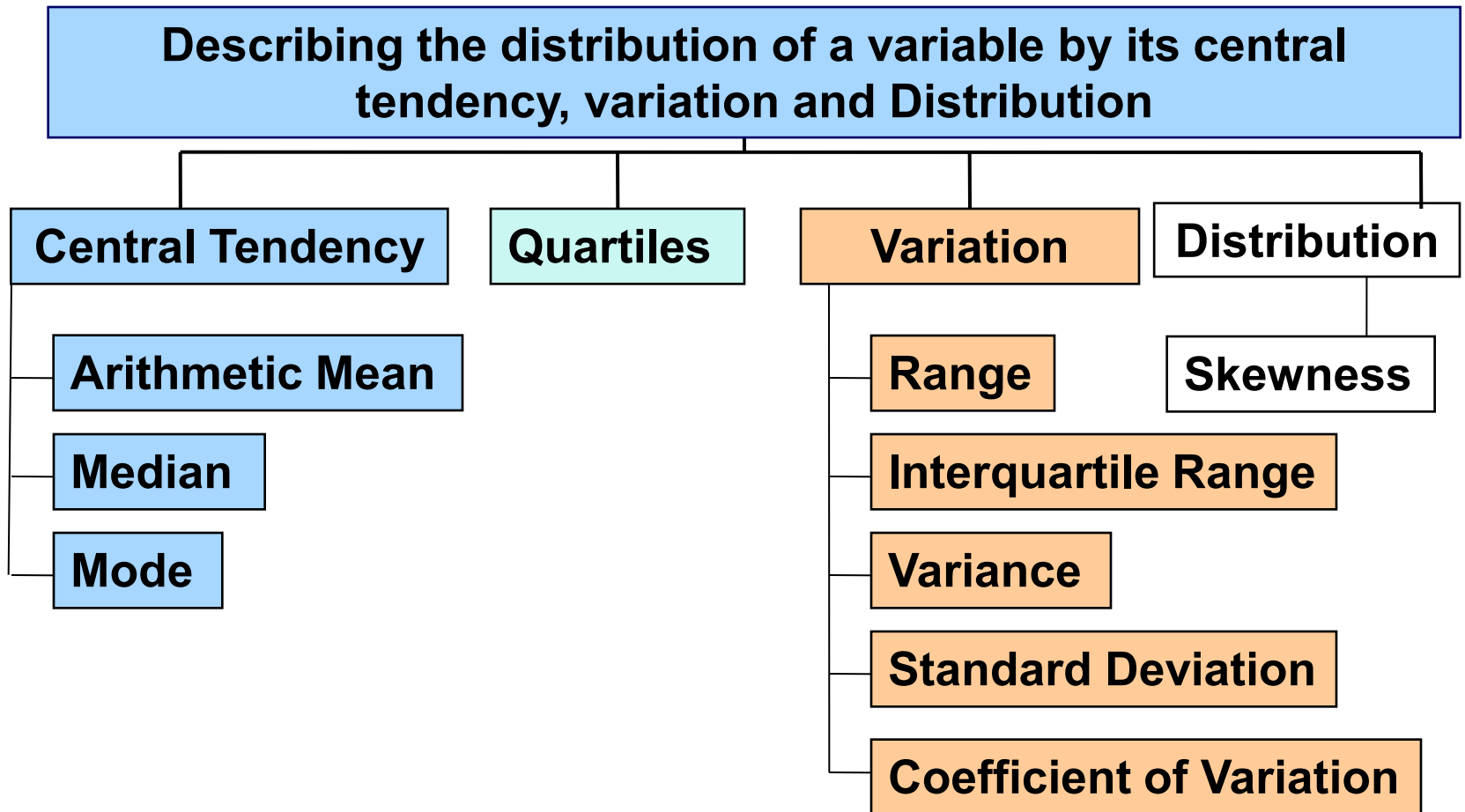
| Year | mean(bmi) |
|------|-----------|
| 2006 | 26.19     |
| 2007 | 26.30     |
| 2008 | 26.33     |
| 2009 | 26.52     |
| 2010 | 26.63     |
| 2011 | 26.56     |
| 2012 | 26.63     |
| 2013 | 26.73     |
| 2014 | 26.77     |
| 2015 | 26.89     |
| 2016 | 27.03     |
| 2017 | 27.14     |
| 2018 | 27.26     |
| 2019 | 27.33     |
| 2020 | 27.55     |
| 2021 | 27.67     |
| 2022 | 27.72     |



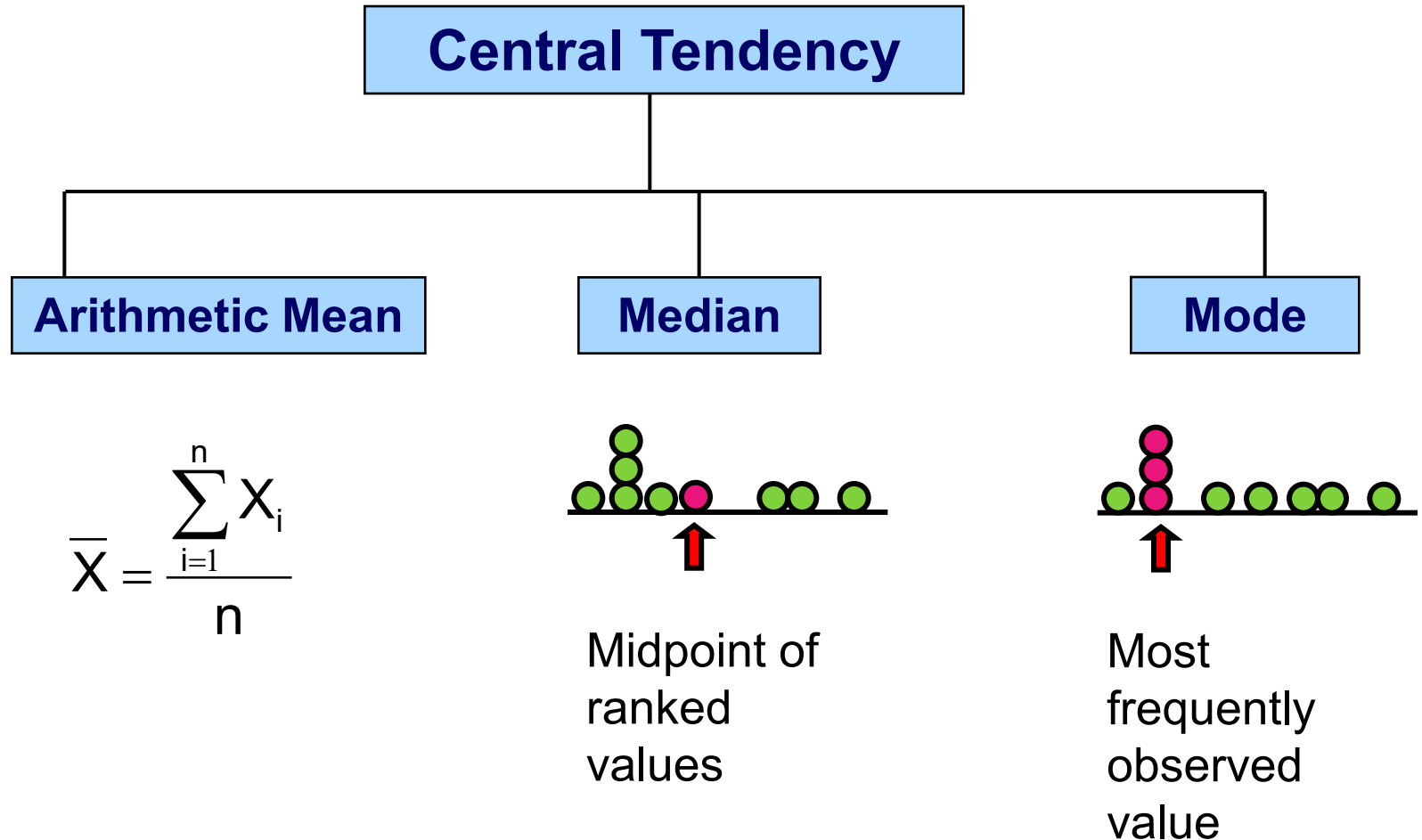
## 6. Graphically represent the Health – BMI relationship and interpret: Cross-section.



# Statistics that describe the Distribution of a Variable

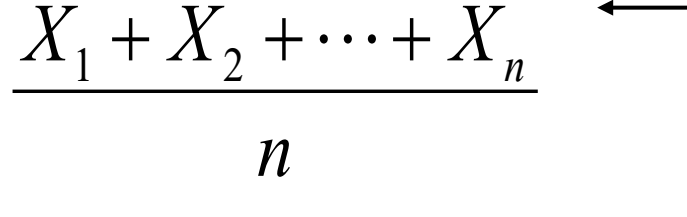


# Measures of Central Tendency



# Arithmetic Mean

For sample of size  $n$  the sample mean is denoted  $\bar{X}$  is calculated:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$


where  $\Sigma$  means “sum” or “add up”

$X_i$ s are all observed values

**Task: Some are scared by mathematical formulas;  
How can we make them easy to understand?**

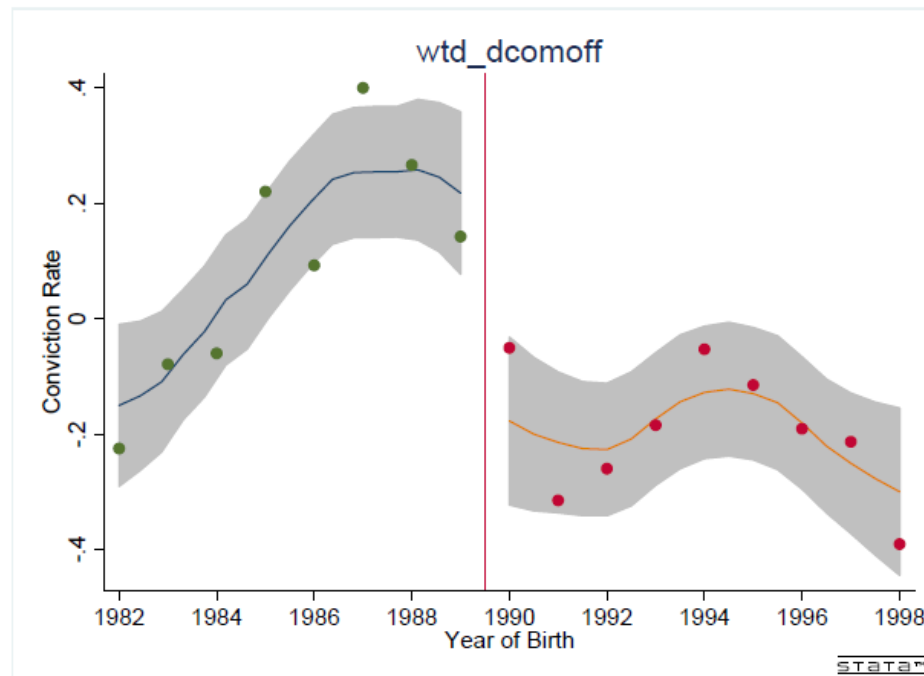
# Excel

| Temp (x) | Soft Drinks (y) |
|----------|-----------------|
| 14       | 60              |
| 22       | 40              |
| 28       | 100             |
| 30       | 100             |
| 36       | 75              |

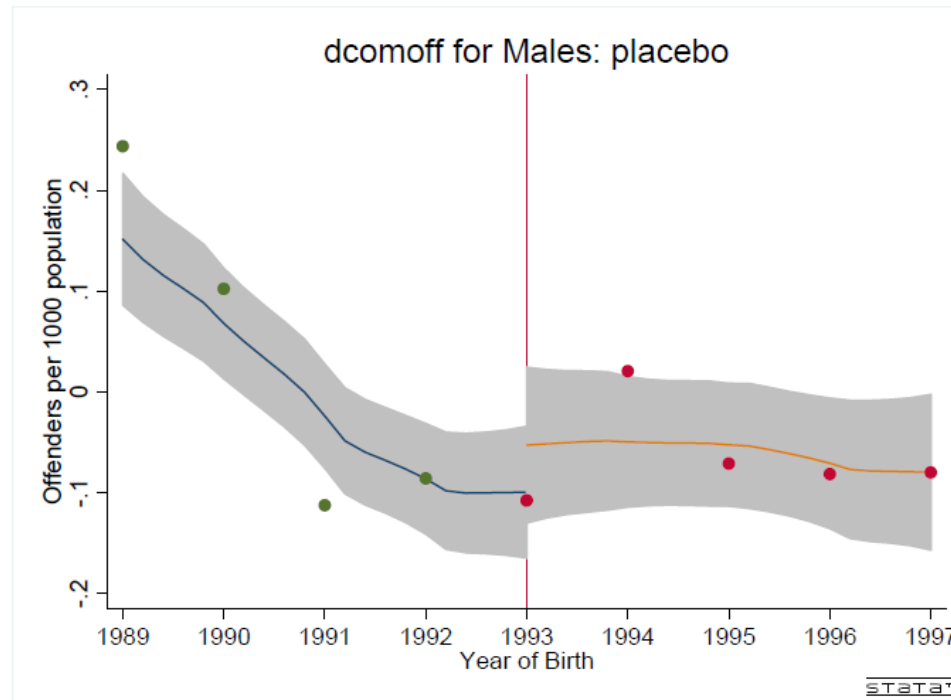
  

|           |  |                                                           |   |              |             |
|-----------|--|-----------------------------------------------------------|---|--------------|-------------|
| Average = |  | $S_x = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$ | = | <b>26.00</b> | =AVG(A3:A7) |
|-----------|--|-----------------------------------------------------------|---|--------------|-------------|

# Example of evidence of causation just using average Education Reform – All Crimes, Male Offender Rate/1000

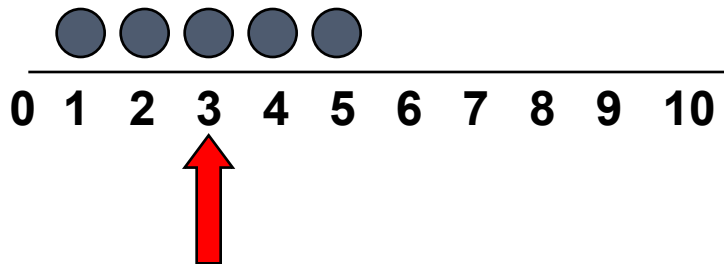


# Example of counterfactual evidence (Placebo) just using average Education Reform – All Crimes, Male Offender Rate/1000: 1993 Placebo



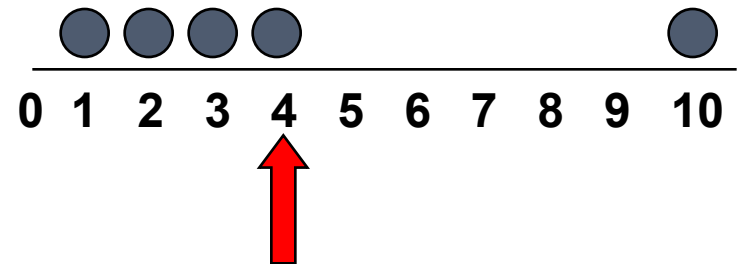
# Arithmetic Mean & Outliers

- The most common measure of central tendency (typically referred to as the mean)
- Commonly called the average
- Calculated as the sum of values divided by the number of values
- Affected by extreme values (outliers)



**Mean = 3**

$$\frac{1 + 2 + 3 + 4 + 5}{5} = \frac{15}{5} = 3$$

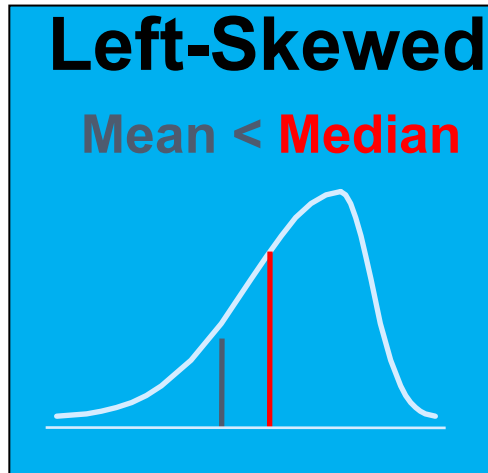


**Mean = 4**

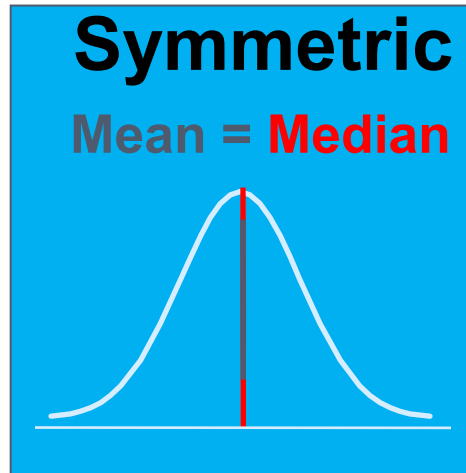
$$\frac{1 + 2 + 3 + 4 + 10}{5} = \frac{20}{5} = 4$$

# Outliers

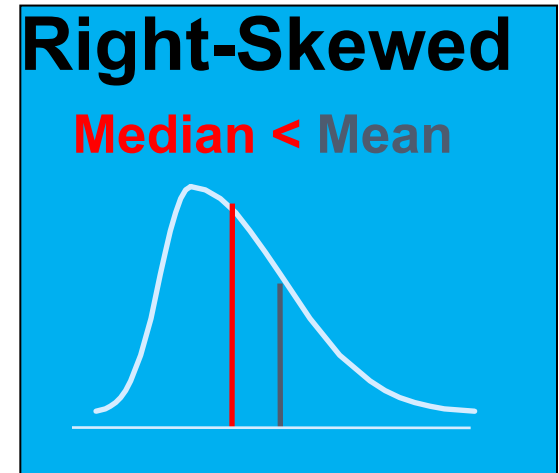
Recall: Shape(s) of a Distribution



Outliers to the left?



No Outliers

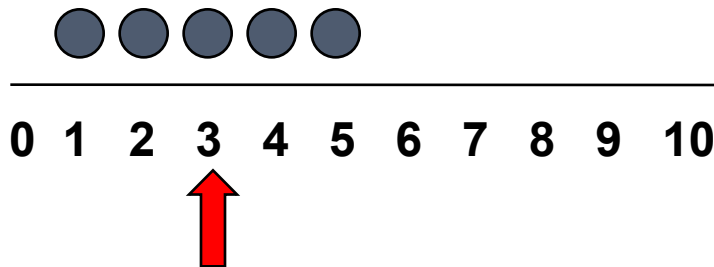


Outliers to the right?

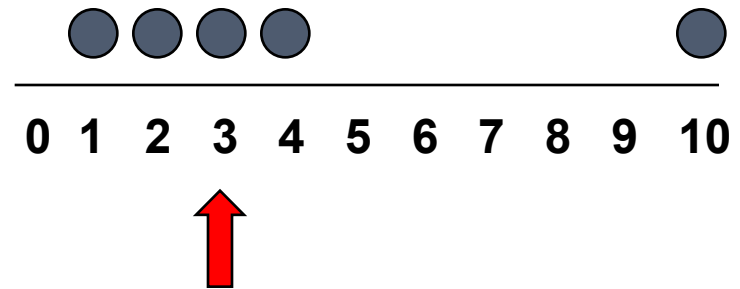
**Task: Why would a distribution be shaped like this?**

# Median

- In an ordered array, the median is the 'middle' number
  - 50% above, 50% below



**Median = 3**



**Median = 3**

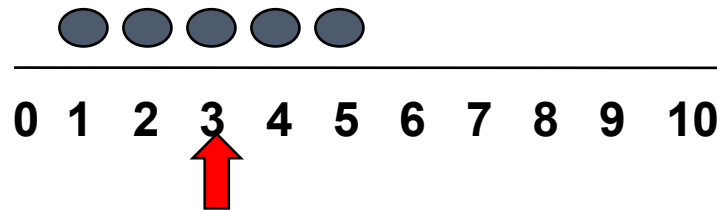
- Not affected by extreme values

# Finding the Median

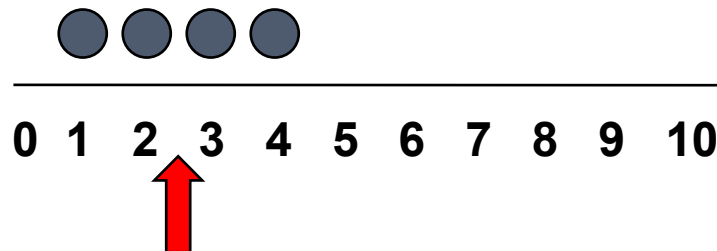
- The location of the median:

$$\text{Median} = \frac{n+1}{2} \text{ ranked value}$$

- Note that  $\frac{n+1}{2}$  is not the *value* of the median, only the *position* of the median in the ranked data
- Rule 1: If the number of values in the data set is odd, the median is the middle ranked value



- Rule 2: If the number of values in the data set is even, the median is the mean (average) of the two middle ranked values



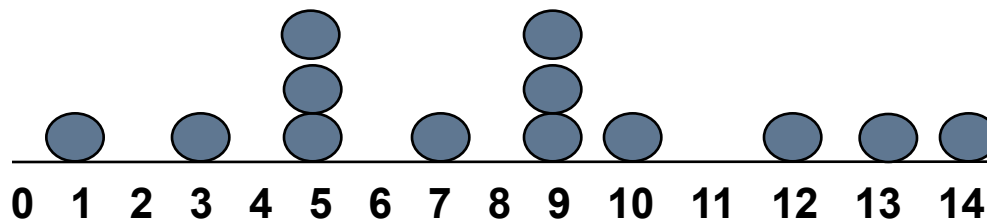
$$\frac{2+3}{2} = 2.5$$

## Excel

# Mode

- A measure of central tendency
- Value that occurs most often (the most frequent)
- Not affected by extreme values
- Unlike mean and median, there may be no unique (single) mode for a given data set
- Used for either numerical or categorical (nominal) data

- There may may be no mode
- There may be several modes



**Modes = 5 and 9**

# Excel

| Temp (x) | Soft Drinks (y) |
|----------|-----------------|
| 14       | 60              |
| 22       | 40              |
| 28       | 100             |
| 30       | 100             |
| 36       | 75              |

|                    |   |        |              |
|--------------------|---|--------|--------------|
| Mode (Soft Drinks) | = | 100.00 | =MODE(B3:B7) |
|--------------------|---|--------|--------------|

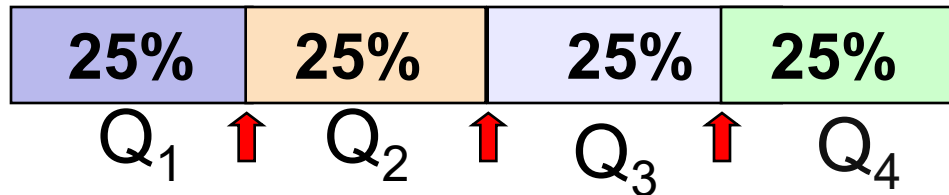
# Quartiles

Quartiles split the ranked data into 4 segments with an equal number of values per segment

The first quartile,  $Q_1$ , is the value for which 25% of the observations are smaller and 75% are larger

The second quartile,  $Q_2$ , is the same as the median (50% are smaller, 50% are larger)

Only 25% of the observations are greater than the third quartile  $Q_3$



**$Q_1$  and  $Q_3$  are measures of non-central location**

**$Q_2$  = median, a measure of central tendency**

# Quartile Formulas

Find a quartile by determining the value in the appropriate position in the *ranked* data, where

First quartile position:  $Q_1 = (n+1)/4$

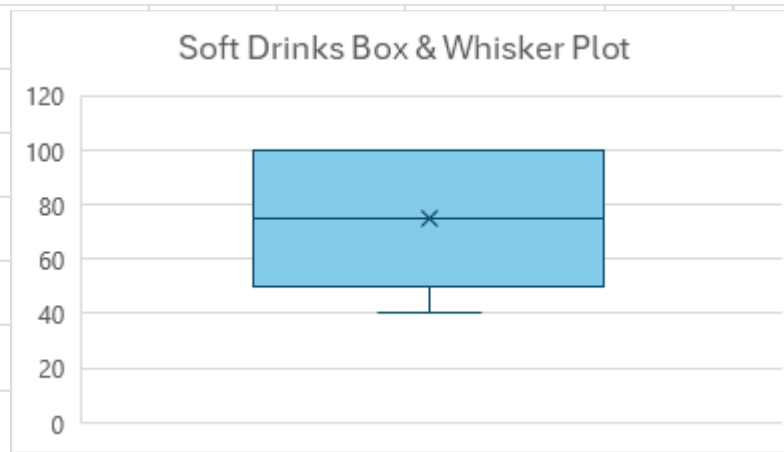
Second quartile position:  $Q_2 = (n+1)/2$  (the median)

Third quartile position:  $Q_3 = 3(n+1)/4$

where **n** is the number of observed values

# Excel

| Temp (x) | Soft Drinks (y) |
|----------|-----------------|
| 14       | 60              |
| 22       | 40              |
| 28       | 100             |
| 30       | 100             |
| 36       | 75              |



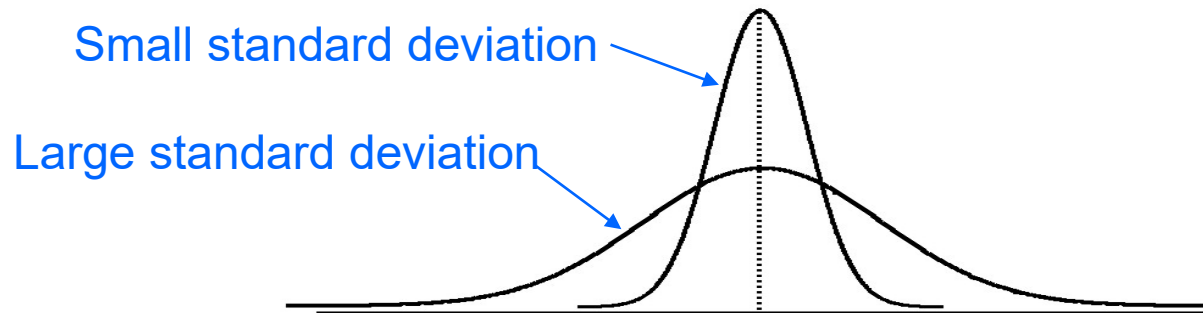
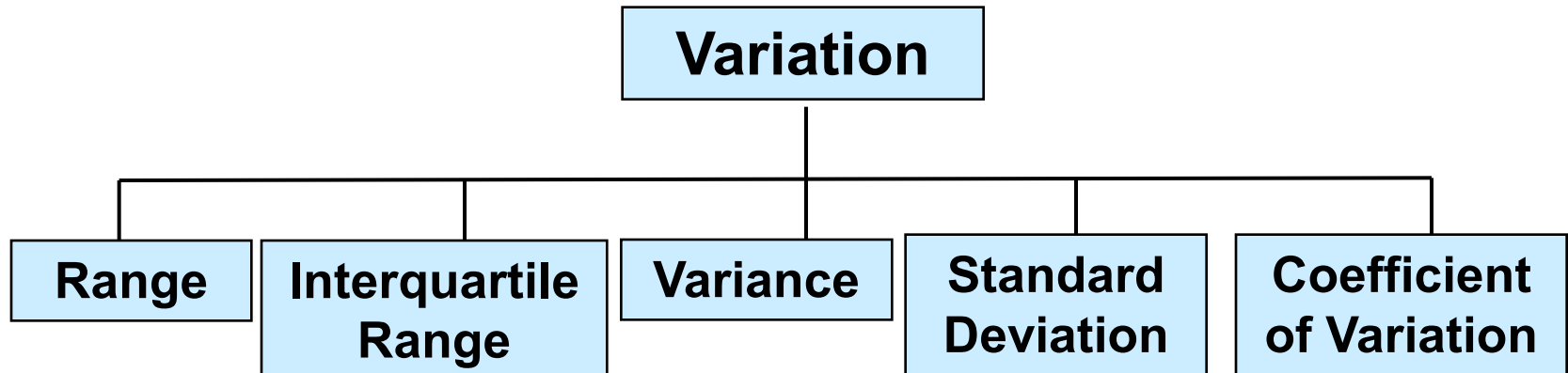
Quartile (Soft Drinks), 4

=

**100**

**=QUARTILE(B3:B7,4)**

# Statistics that describe Variation in the Distribution of a Variable



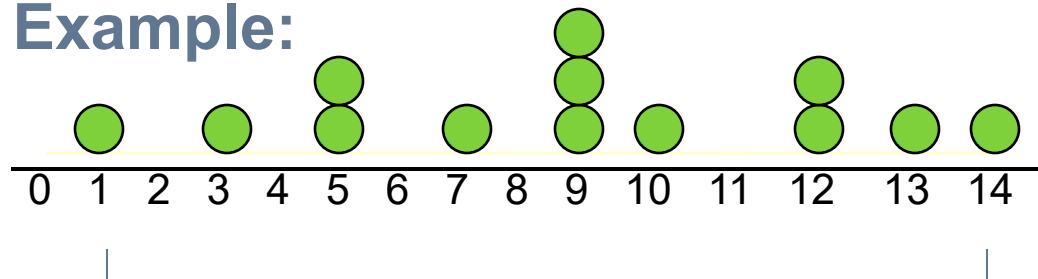
[Click here to view Variation in a Distribution](#)

# Range

- Simplest measure of variation
- Difference between the largest and the smallest values in a set of data

$$\text{Range} = X_{\text{largest}} - X_{\text{smallest}}$$

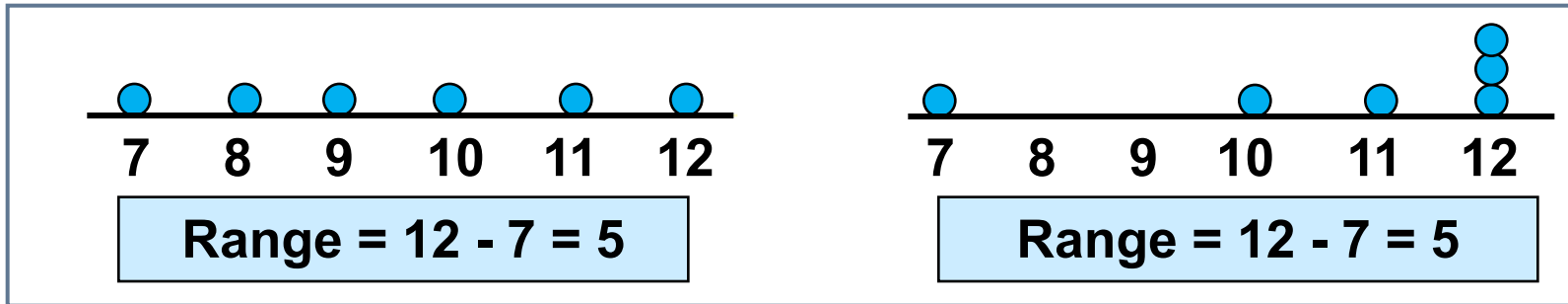
Example:



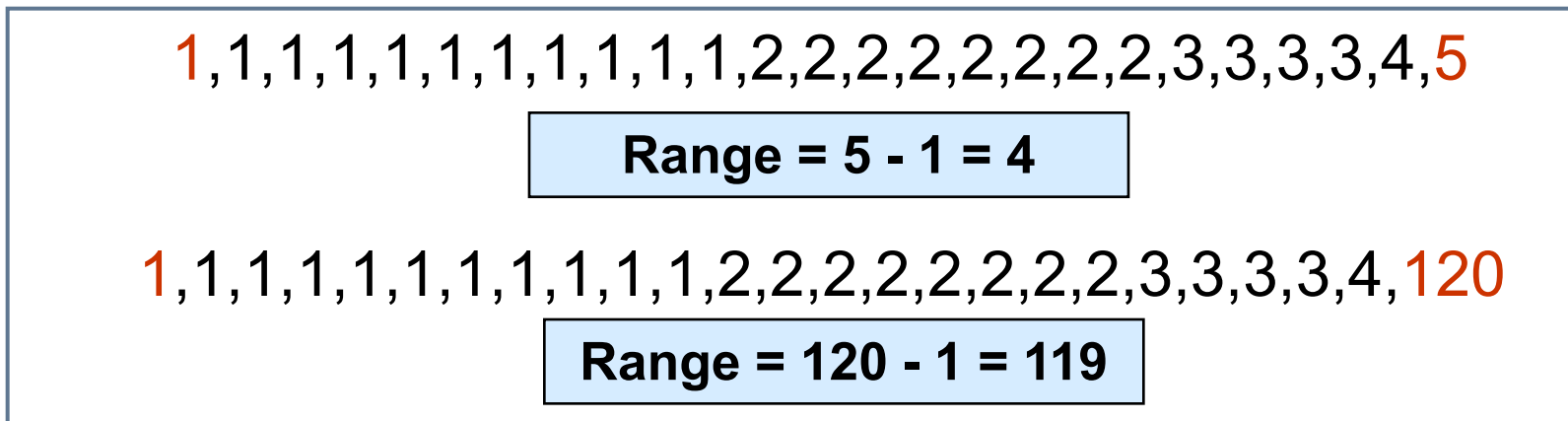
$$\text{Range} = 14 - 1 = 13$$

# Disadvantages of the Range

- Ignores the distribution of the data



- Like the mean, range is sensitive to outliers



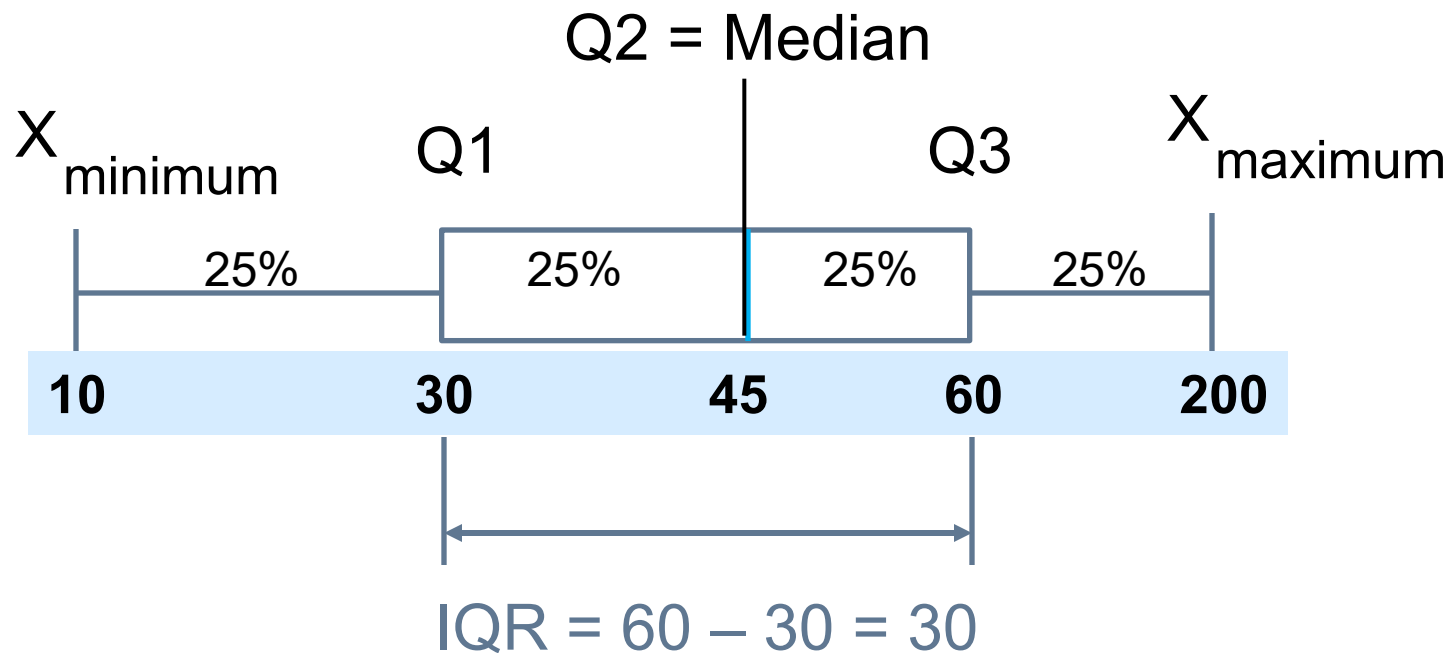
# Interquartile Range (IQR)

- Like the median and  $Q_1$  and  $Q_2$ , the IQR is a resistant summary measure
- Eliminates outlier problems by using the **interquartile range** as high- and low-valued observations are removed from calculations
- $\text{IQR} = 3^{\text{rd}} \text{ quartile} - 1^{\text{st}} \text{ quartile}$

$$Q_3 - Q_1$$

# Interquartile Range

Example: Range =  $200 - 10 = 190$  (Misleading)



Even if the value of 200 changes to 300, IQR remains the same, hence resistant to changes in extreme values

# Coefficient of Variation

- Measures relative variation i.e. shows variation relative to mean
- Can be used to compare two or more sets of data measured in different units
- Always expressed as percentage (%)

$$CV = \left( \frac{S}{\bar{X}} \right) * 100\%$$

# Excel

| Temp (x) | Soft Drinks (y) |
|----------|-----------------|
| 14       | \$ 60.00        |
| 22       | \$ 40.00        |
| 28       | \$ 100.00       |
| 30       | \$ 100.00       |
| 36       | \$ 75.00        |

|                                          |                                                 |   |              |   |                           |
|------------------------------------------|-------------------------------------------------|---|--------------|---|---------------------------|
| Coefficient of Variation (Temp) =        | $CV = \left( \frac{S}{\bar{X}} \right) * 100\%$ | = | <b>32.18</b> | % | =STDEV(A3:A7)/=AVG(A3:A7) |
| Coefficient of Variation (Soft Drinks) = |                                                 | = | <b>34.64</b> | % | =STDEV(B3:B7)/=AVG(B3:B7) |

# Dispersion of Variables about their Averages

- Population
- Sample

# Dispersion of the variables about their averages: Variance

- Mean squared deviation

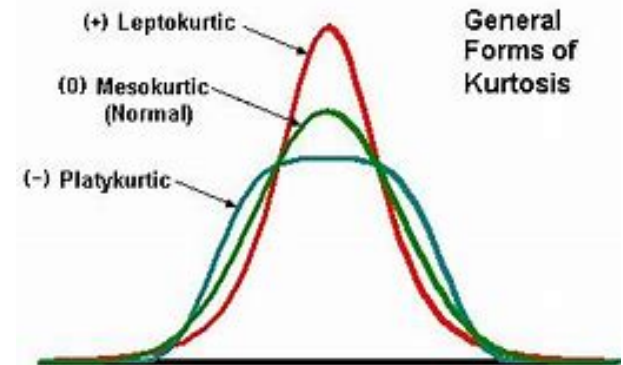
Sample Variance

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

The denominator  $(n-1)$  is to adjust for the biasness of the sample statistics.

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$



**Task: Why are the variance formulas different?**

# Variance Example

|         |   |   |   |   |   |   |   |   |   |   |
|---------|---|---|---|---|---|---|---|---|---|---|
| Stock A | 0 | 2 | 5 | 5 | 5 | 6 | 6 | 7 | 8 | 9 |
| Stock B | 4 | 4 | 5 | 5 | 5 | 5 | 6 | 6 | 6 | 7 |

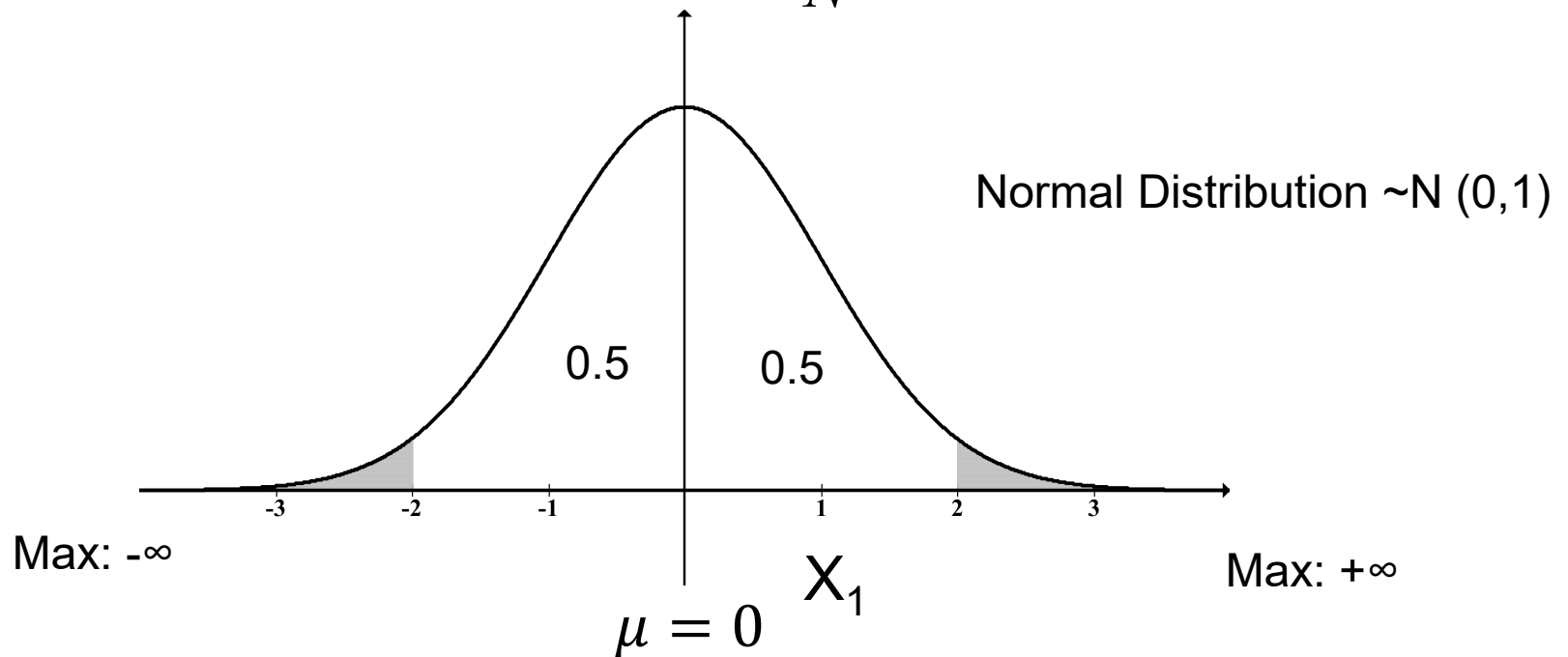
Calculate the sample standard deviation for stocks A and B

$$\begin{aligned}s_A^2 &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} \\&= \frac{(0 - 5.3)^2 + (2 - 5.3)^2 + \dots + (9 - 5.3)^2}{10 - 1} \\&= \frac{64.1}{9} \\&= 7.12 \\s_A &= \sqrt{7.12} \\&= 2.67\end{aligned}$$

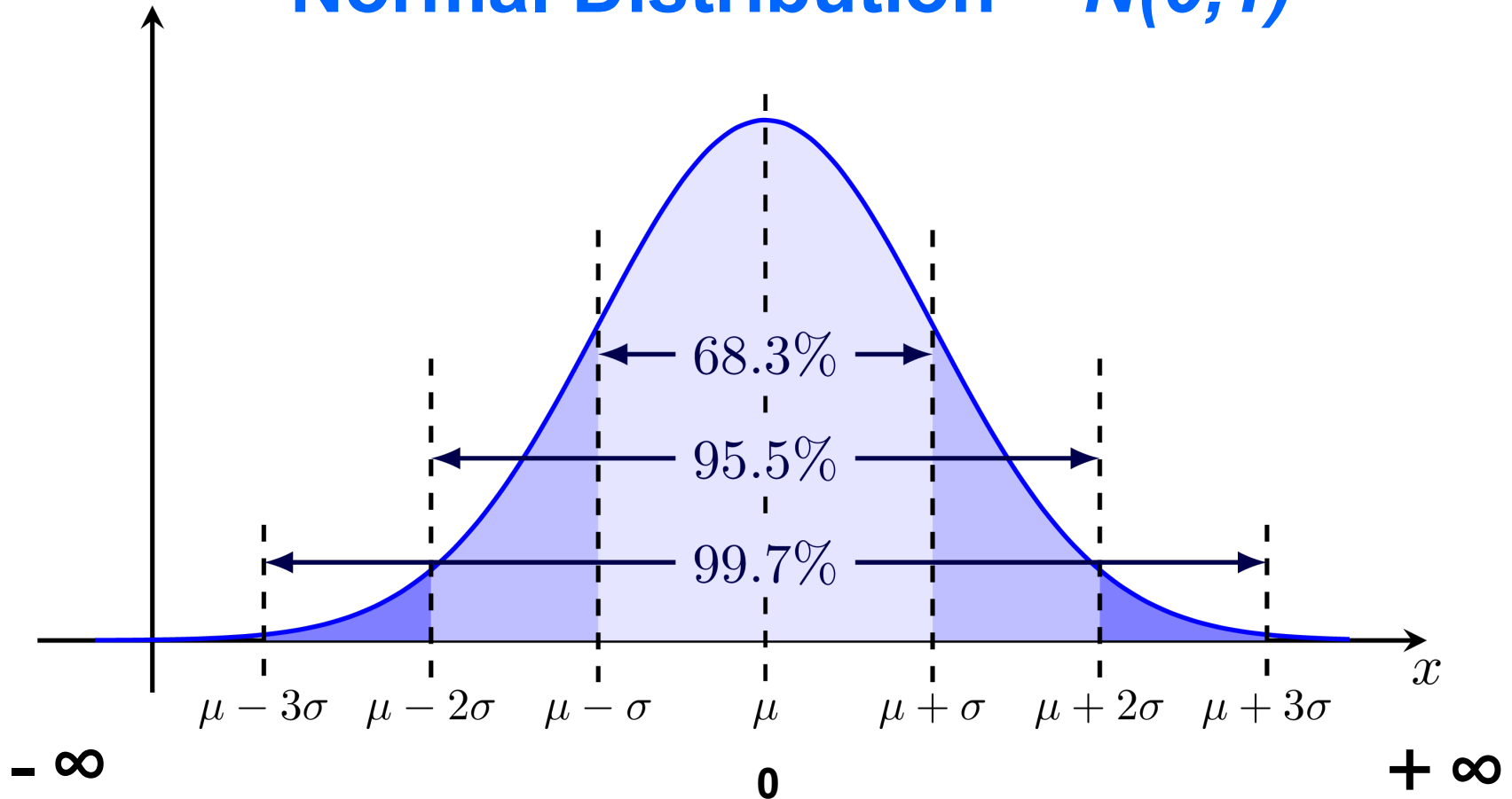
# Standard Deviation - Population

Most commonly used measure of variation; Shows variation about the mean; has the same units as the original data

$$\sigma = \sqrt{\sigma^2}; \quad \sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N} \quad ; (+0 \dots +\infty)$$



# Dispersion of $X$ values in a Normal Distribution $\sim N(0,1)$



e.g. 68.3% of observations lie within  $\pm 1$  Standard Deviation, either side, of the mean

# Standard Deviation - Sample

Most commonly used measure of variation;  
Shows variation about the mean; has the  
same units as the original data

$$S = \sqrt{S^2}; \quad S_x = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

# Excel

| Temp (x) | Soft Drinks (y) |
|----------|-----------------|
| 14       | 60              |
| 22       | 40              |
| 28       | 100             |
| 30       | 100             |
| 36       | 75              |

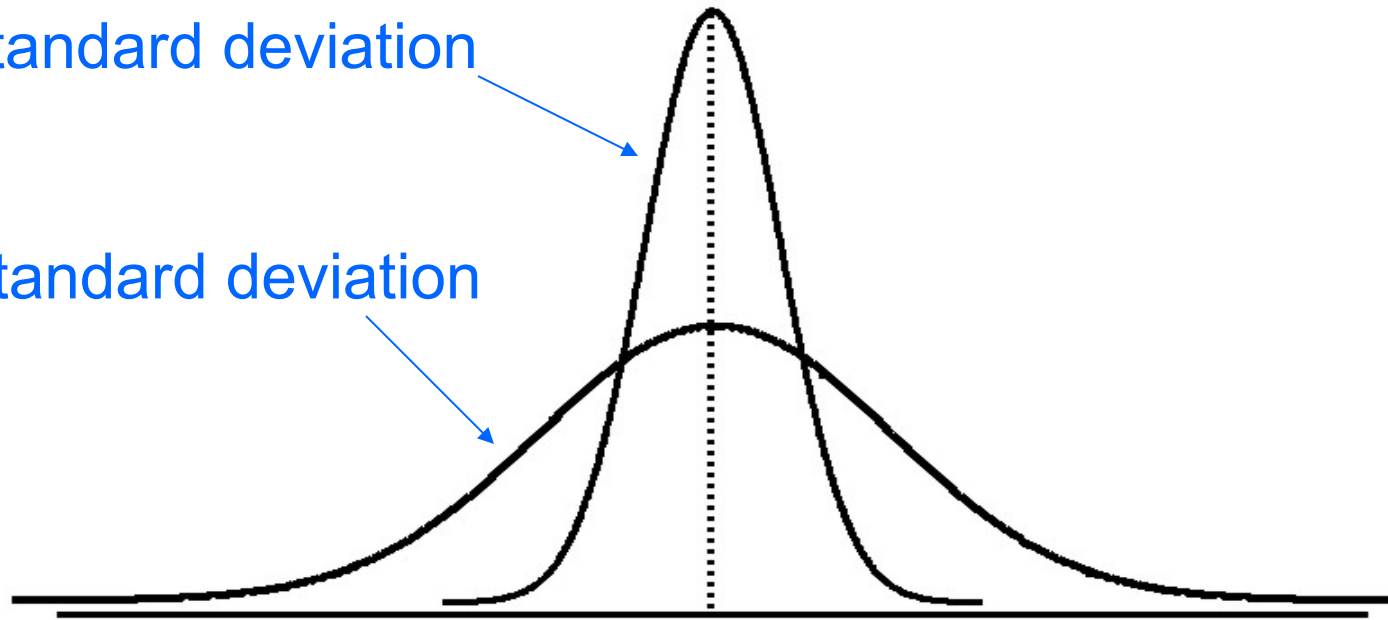
  

|                      |                                                           |   |             |                 |
|----------------------|-----------------------------------------------------------|---|-------------|-----------------|
| Standard Deviation = | $S_x = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$ | = | <b>8.37</b> | =STDEV.S(A3:A7) |
|----------------------|-----------------------------------------------------------|---|-------------|-----------------|

# Visualize a distribution

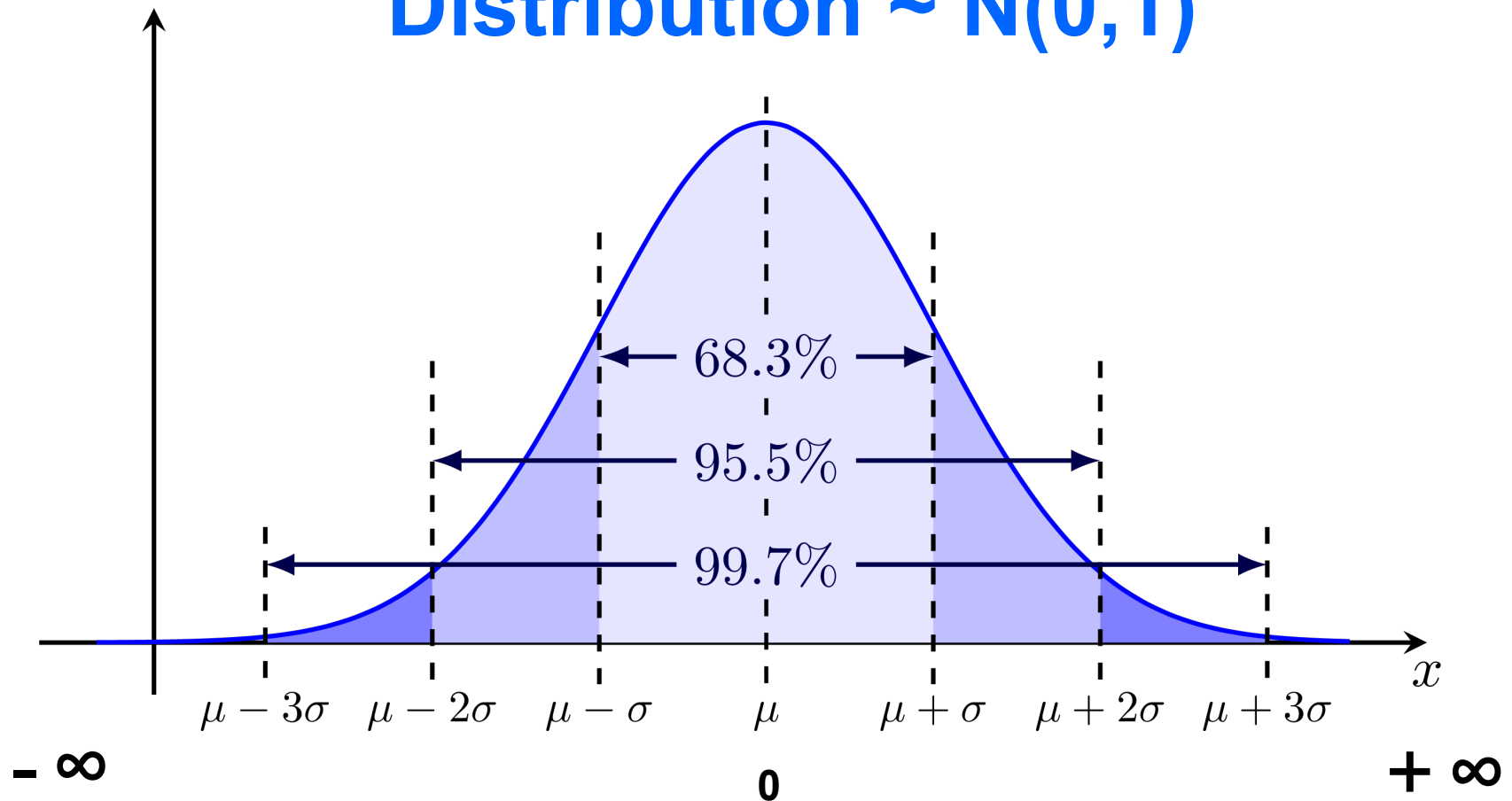
Small standard deviation

Large standard deviation



# Continuous Probability Distributions

# Dispersion of X values in a Normal Distribution $\sim N(0,1)$



e.g. 68.3% of observations lie within +/- 1 Standard Deviation, either side, of the mean

# Distance from the Mean - Population

We can also express the position of a value of  $x$  relative to its mean in terms of a **continuous probability**

Normal Distribution:  $\sim N(0, 1)$

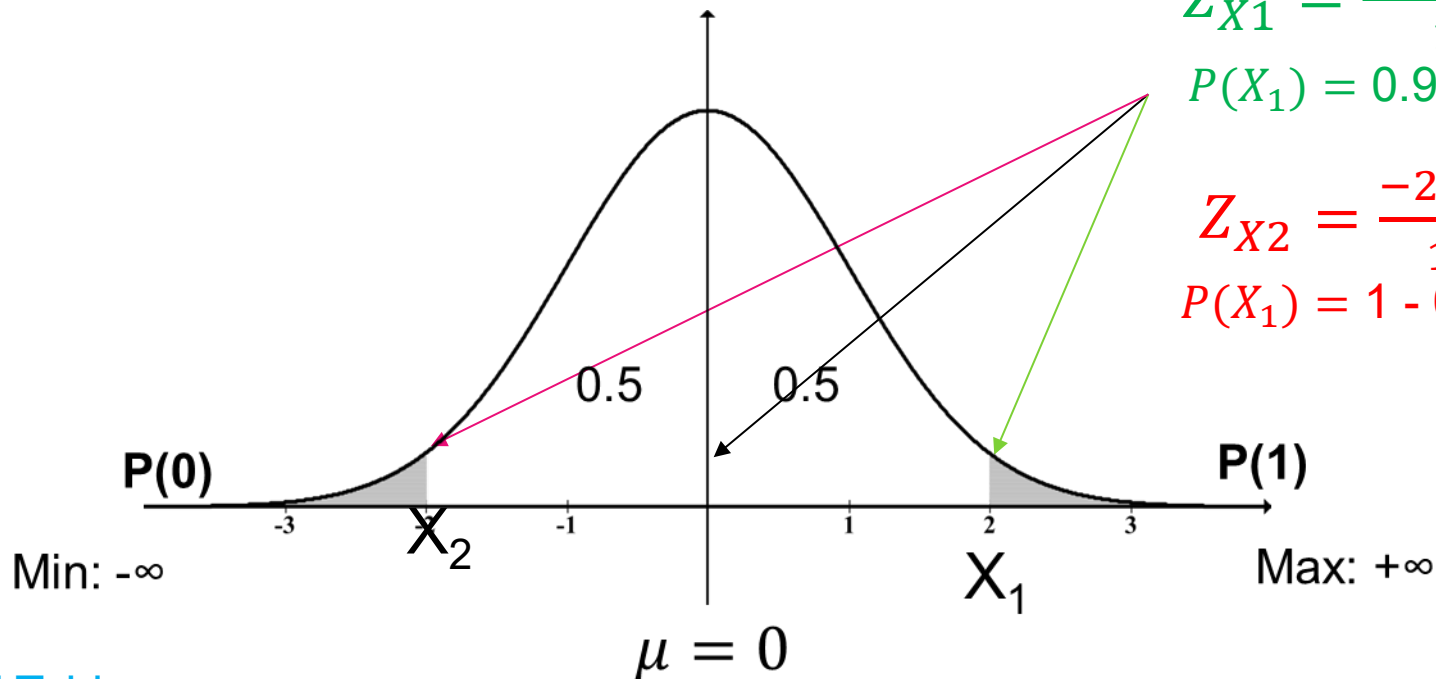
$$Z = \frac{(x_i - \mu)}{\sigma}$$

$$Z_{X_1} = \frac{(2-0)}{1} = 2$$

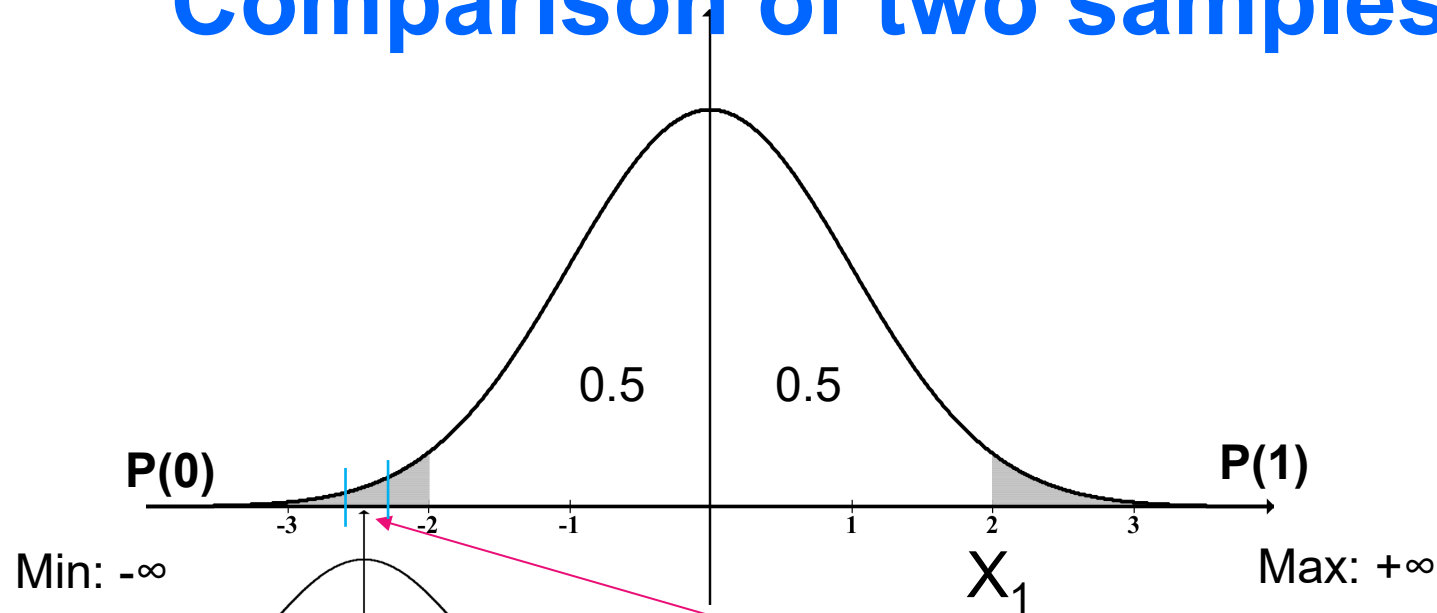
$$P(X_1) = 0.9772$$

$$Z_{X_2} = \frac{-2-0}{1} = -2$$

$$P(X_1) = 1 - 0.9772 = 0.0228$$

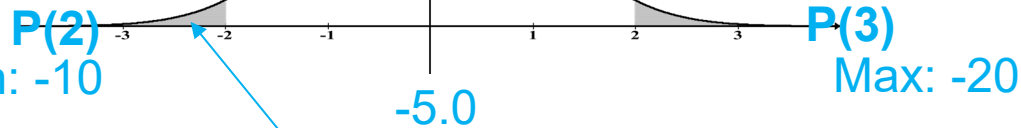


# Break the distribution into classes: Comparison of two samples



$$Z_{X_2} = \frac{-2-0}{1} = \underline{-2}$$

$$P(X_1) = 1 - 0.9772 = \underline{0.0228}$$



$$Z_{X_3} = \frac{-7.5-5}{1.25} = \underline{-2}$$

$$P(X_3) = 1 - 0.9772 = \underline{0.0228}$$

# Next Geek 2: Relationship Statistics

End of Session